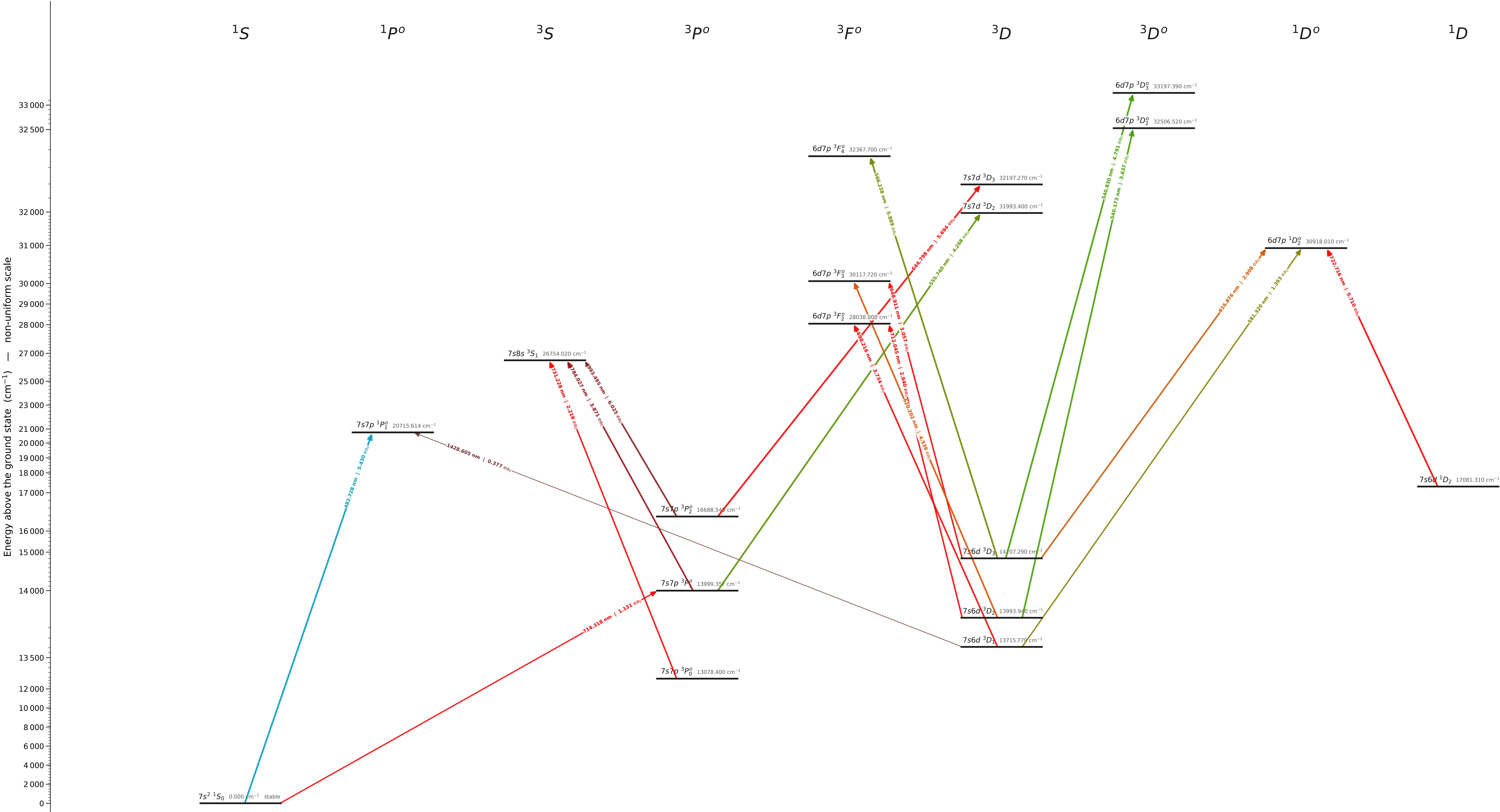
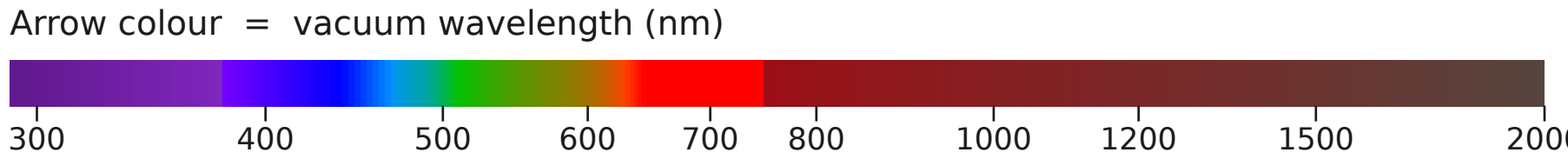


Radium

Ra (Ra I). Strongest classified transitions below 2 μm (NIST): 18 levels, 18 lines.

Ra⁺ ionisation limit 42573.36 cm⁻¹ = 5.27842 eV (234.889 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U\|er\|J\rangle|$ in e a₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U\ er\ J\rangle $ (e a ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
7s ² 1S ₀ - 7s7p 1P ^o ₁	482.7277	482.5928	621.0385	5.430	1.770e+08	-	-	
7s6d 3D ₂ - 6d7p 3D ^o ₂	540.1732	540.0231	554.9932	3.637	3.400e+07	-	-	
7s6d 3D ₃ - 6d7p 3D ^o ₃	540.8300	540.6796	554.3193	4.791	4.200e+07	-	-	
7s7p 3P ^o ₁ - 7s7d 3D ₂	555.7395	555.5852	539.4478	4.268	4.300e+07	-	-	
7s6d 3D ₃ - 6d7p 3F ^o ₃	566.2383	566.0812	529.4458	5.889	4.300e+07	-	-	
7s6d 3D ₁ - 6d7p 1D ^o ₂	581.3199	581.1588	515.7099	1.393	4.000e+06	-	-	
7s6d 3D ₃ - 6d7p 1D ^o ₂	616.8757	616.7051	485.9852	2.908	1.460e+07	-	-	
7s6d 3D ₂ - 6d7p 3F ^o ₃	620.2020	620.0304	483.3788	4.539	2.500e+07	-	-	
7s7p 3P ^o ₂ - 7s7d 3D ₃	644.7981	644.6200	464.9400	5.694	3.500e+07	-	-	
7s6d 3D ₃ - 6d7p 3F ^o ₃	648.9112	648.7319	461.9931	3.057	9.900e+06	-	-	
7s6d 3D ₁ - 6d7p 3F ^o ₃	698.2157	698.0232	429.3694	3.734	1.660e+07	-	-	
7s6d 3D ₂ - 6d7p 3F ^o ₃	712.0448	711.8486	421.0303	2.940	9.700e+06	-	-	
7s6d 1D ₂ - 6d7p 1D ^o ₂	722.7157	722.5165	414.8138	5.710	3.500e+07	-	-	
7s7p 3P ^o ₀ - 7s8s 3S ₁	731.2283	731.0269	409.9848	2.218	8.500e+06	-	-	
7s7p 3P ^o ₂ - 7s8s 3S ₁	784.0270	783.8113	382.3752	3.871	2.100e+07	-	-	
7s7p 3P ^o ₂ - 7s8s 3S ₁	993.4946	993.2223	301.7555	6.025	2.500e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).